A3-PS013-TC452

Incident Timeline and Evidence Collection for Small Teams

TEAM EXPOSE

Anurag Deb, Sangram Keshari Patra
Corresponding email ID: anuragdeb2005@gmail.com

Mentor: Prof. Partha Chakraborty, IEM Kolkata

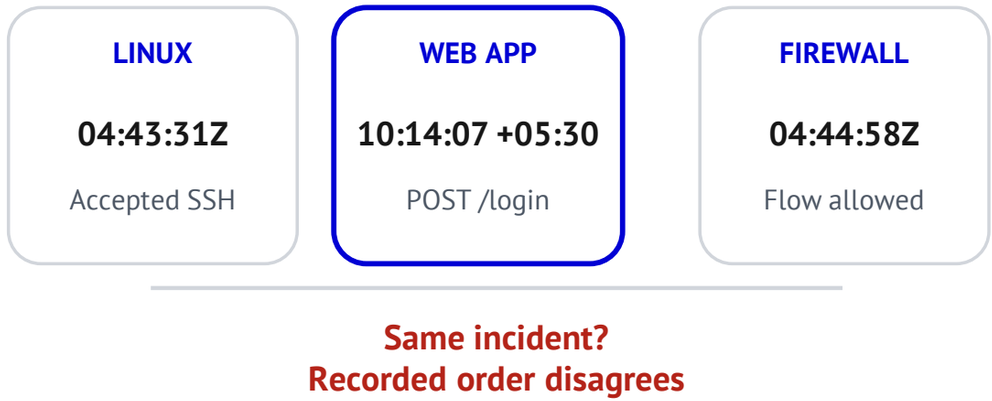
Project: ALIBI

October 11, 2026

Problem Description

A small organization is breached. Its Linux server, web application and firewall each record a fragment - but the formats, time zones and clocks disagree. Some records are missing.

Without a dedicated SOC, an IT administrator must manually decide what happened, in what defensible order, and which evidence can be trusted or handed to an external responder. A simple timestamp sort can create a confident but wrong story.



WHY THIS IS HARD

- **Heterogeneous text logs**
- **Timezone and clock skew**
- **Incomplete or conflicting records**
- **Weak evidence provenance**

RISK OF NAIVE SORTING

Authentication and access may appear to occur before the traffic that supports them. The uncertainty is hidden instead of explained.

PS-013 requires a lightweight, searchable UTC timeline with preserved evidence and exportable reporting.

Background

COMMON SMALL-TEAM INVESTIGATION WORKFLOW



Operational problem

Resource-constrained teams often combine command-line tools, spreadsheets and ad hoc notes. Repeating the work consistently, preserving lineage and explaining assumptions are difficult without a dedicated workflow.

Technical challenge

UTC conversion solves time zones, not an inaccurate clock. If Host A is 73 s slow and Host B is 88 s slow, sorting their normalized timestamps still misorders events. Corrections need repeated cross-source anchors, residual checks and explicit uncertainty.

Evidence integrity is a separate concern: hashing supports change detection, but does not by itself establish legal chain of custody or forensic certification. [1][2]

Literature & RFC Review

Relevant work establishes the components. The opportunity is a small-team workflow that joins them while keeping uncertainty and provenance visible. An undocumented feature is not treated as proof of absence.

Source	Established contribution	What remains for this proposal
NIST SP 800-86 [1]	Collection, examination, analysis and reporting; evidence integrity guidance	Translate sound practice into a lightweight, repeatable product workflow
NIST SP 800-61r3 [2]	Incident response aligned to cybersecurity risk management	Operational context, not a temporal reconstruction method
Timesketch + Plaso [3][4]	Broad extraction, searchable forensic timelines, collaboration and analyzers	Reviewed docs do not establish ALIBI's integrated uncertainty-aware calibration and claim lineage
Wazuh [5]	Agent/server/indexer/dashboard architecture for collection, analysis and search	A one-off local evidence workbench has a different operating model
Lamport [6]	Partial ordering of distributed events without assuming one perfect global clock	Adapt the principle to human-verifiable incident reconstruction
RFC 5848 [7]	Signed syslog messages and integrity concepts	MVP uses SHA-256 verification; signing remains a future extension
Chronos prototype [8]	Hashes, multi-format parsing, skew inference, timelines and graphs	Prevents a novelty claim based on those features alone; ALIBI focuses on intervals, challenge checks and claim-to-byte verification

Proposed Solution

ALIBI: A Proposed Evidence-First Incident Reconstruction Framework

A lightweight, local, offline-first workbench for reconstructing incidents from heterogeneous logs without inventing certainty.



PROPOSED INTEGRATION

ALIBI does not claim new underlying algorithms. Its defensible contribution is the integrated audit path: original evidence → supported clock model → interval-aware events → typed relationships → challenged conclusions → independently verifiable export.

MVP: stages 1-5 with deterministic rules and no paid API. Later: cryptographic signing and tightly constrained narrative assistance.

Optimization Proposed

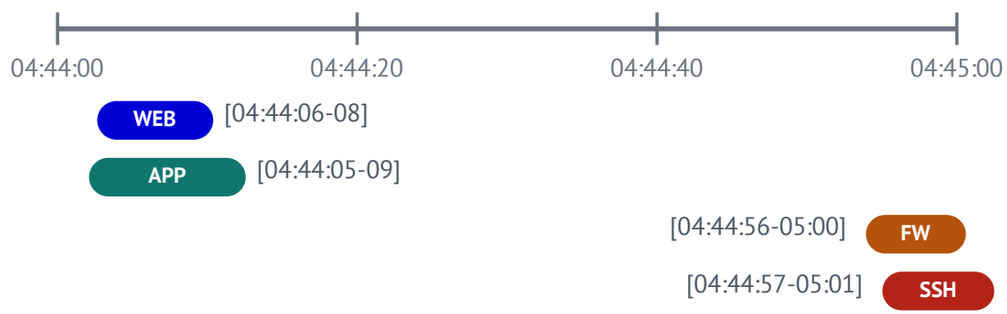
Synthetic worked example - values are test fixtures, not measured performance.

BEFORE: NAIVE UTC SORT

1	APP	04:42:54Z	auth.success R7
2	SSH	04:43:31Z	Accepted :49812
3	WEB	04:44:07Z*	POST /login R7
4	FW	04:44:58Z	Allow :49812

* Web 10:14:07 +05:30. The sort wrongly places app and SSH activity before supporting traffic.

AFTER: SUPPORTED CALIBRATION



Repeated request-ID / unique flow anchors support +73 s (app) and +88 s (SSH). One match is insufficient; investigator approval is required.

Confident: WEB → APP; APP before SSH cluster. Uncertain: FW vs SSH order. Gap: no network records 04:45:10-04:52:10.

Conventional approach	ALIBI proposed optimization
Sort by normalized time	Calibrate from repeated anchors
Assume exact event time	Keep corrected time intervals
Treat proximity as a link	Require typed evidence edges
Review silence/conflict manually	Flag gaps and contradictions
Separate report and source	Trace claim to record and hash

Solution Architecture & Design

Local system boundary, components and implementation choices

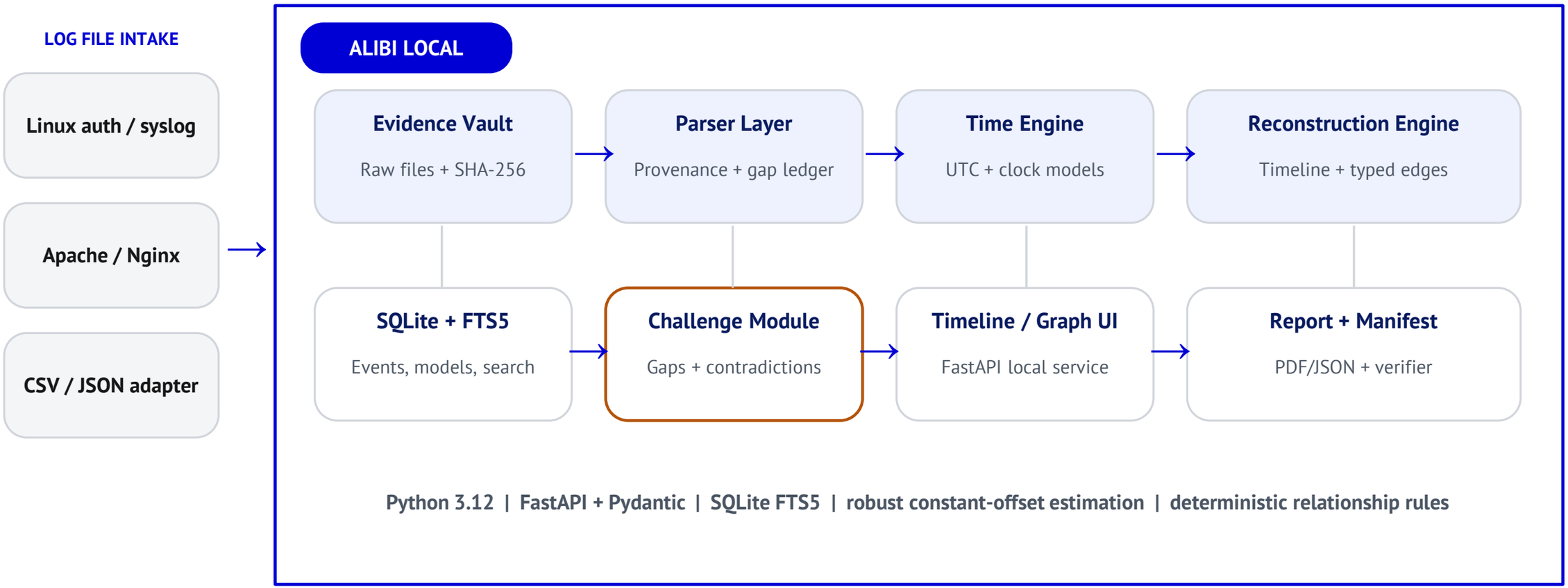
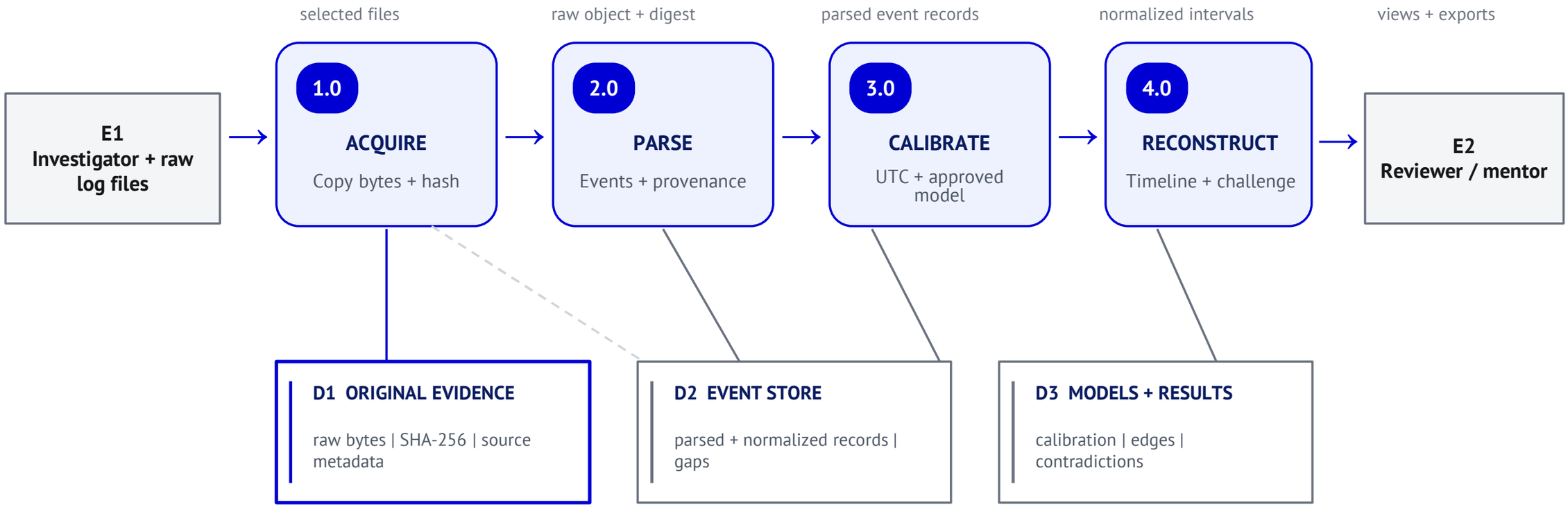


Figure: Proposed ALIBI UML component architecture. All core processing remains local and offline.

Data Flow Diagram

Data movement is shown separately from system components. Original evidence is written once and never overwritten.



Flow control: proposed clock models are reviewed before application. Derived records reference D1, so every material claim can be traced to a raw byte/line range and verified hash.

Figure: ALIBI level-1 data flow diagram.

Deployment Timeline

Planned prototype roadmap - target technical showcase: October 29, 2026

OCT 12-15

FOUNDATION

Evidence intake; two parsers; SHA-256; original evidence storage

OCT 16-19

REQUIRED TIMELINE

UTC normalization; manual offset support; searchable timeline

OCT 20-23

TEMPORAL MODEL

Anchor-based calibration; corrected intervals; basic reconstruction rules

OCT 24-26

CHALLENGE + PROOF

Gaps/contradictions; evidence-linked report; three synthetic datasets

OCT 27-28

VALIDATE

Tests; manual triage comparison; demo preparation

OCT 29

SHOWCASE

Target technical prototype demonstration

MVP gate: satisfy every mandatory PS-013 requirement before optional signing, AI narration or additional parsers.

Bibliography I

[1] K. Kent et al., Guide to Integrating Forensic Techniques into Incident Response, NIST SP 800-86, Aug. 2006. doi:10.6028/NIST.SP.800-86

[2] A. Nelson et al., Incident Response Recommendations and Considerations for Cybersecurity Risk Management, NIST SP 800-61 Rev. 3, Apr. 2025. doi:10.6028/NIST.SP.800-61r3

[3] Timesketch, Collaborative forensic timeline analysis and user documentation. https://timesketch.org/ (accessed 11 Oct. 2026)

[4] Plaso documentation, log2timeline / psort event extraction and processing. https://plaso.readthedocs.io/ (accessed 11 Oct. 2026)

[5] Wazuh documentation, Architecture. https://documentation.wazuh.com/current/getting-started/architecture.html (accessed 11 Oct. 2026)

[6] L. Lamport, Time, Clocks, and the Ordering of Events in a Distributed System, Communications of the ACM, 21(7), 1978. doi:10.1145/359545.359563

[7] J. Kelsey et al., Signed Syslog Messages, RFC 5848, May 2010. https://www.rfc-editor.org/rfc/rfc5848

[8] BaggaByte, Chronos: Automated Multi-Source Forensic Timeline Parser & Incident Reconstruction Engine. https://github.com/BaggaByte/chronos (reviewed 11 Oct. 2026)

[9] AIORI Hackathon 3, PS-013: Incident Timeline and Evidence Collection for Small Teams, official problem statement supplied to Team EXPOSE, 2026.

All URLs and publication details verified against primary documentation available on 11 October 2026.

ALIBI

A Proposed Evidence-First Incident Reconstruction Framework

Proposed by Team EXPOSE

AIORI-3 | PS-013

Thank You for Your Attention!